H_2O

C_{2v}	E	C_2	$\sigma_v(xy)$	$\sigma_d(yz)$
Γ	2	0	2	0

From the character table of C_{2v} we get,

No. of $A_1 = \frac{1}{4}[(2\times1\times1)+0+(2\times1\times1)+0] = 1$

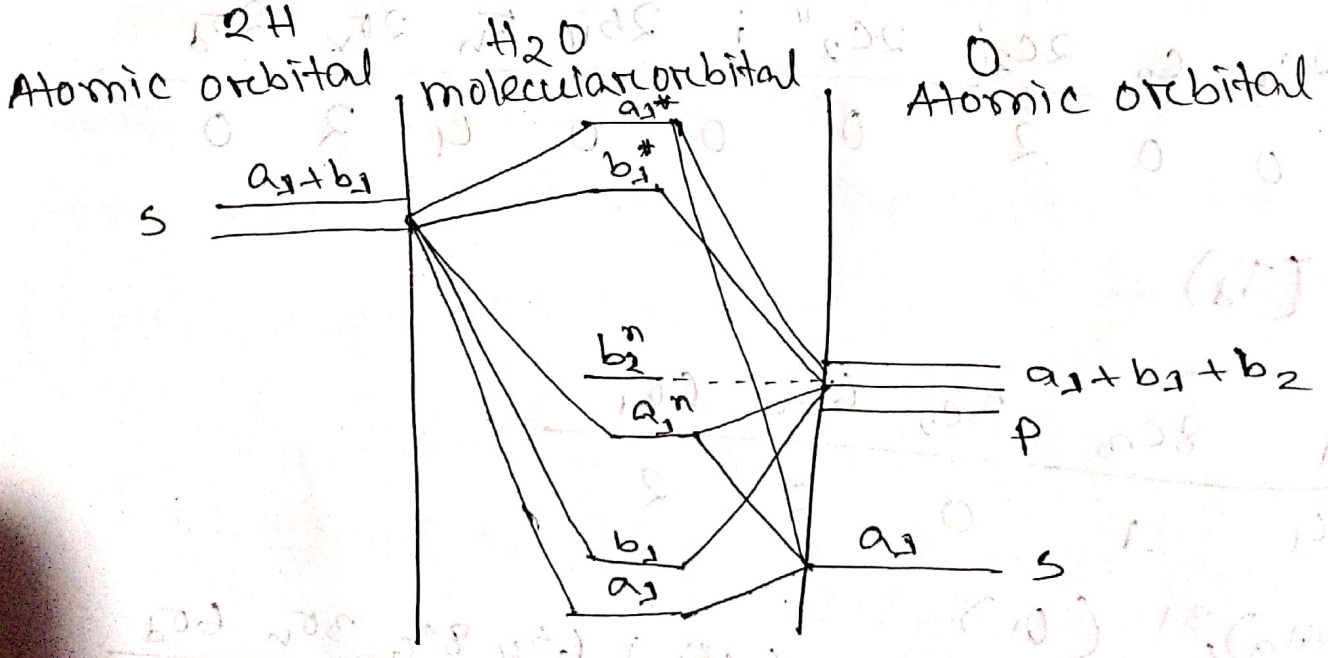
No. of $A_2 = \frac{1}{4}[(2\times1\times1)+0+\{2\times(-1)\times1\}+0] = 0$

No. of $B_1 = \frac{1}{4}[(2\times1\times1)+0+(2\times1\times1)+0] = 1$

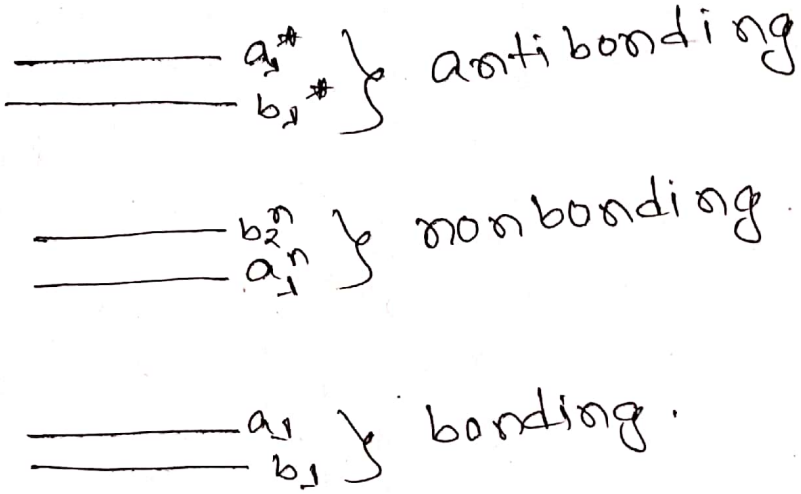
No. of $B_2 = \frac{1}{4}[(2\times1\times1)+0+\{2\times(-1)\times1\}+0] = 0$

So, we get. $\Gamma = A_1 + B_1$.

Correlation diagram for H_2O.

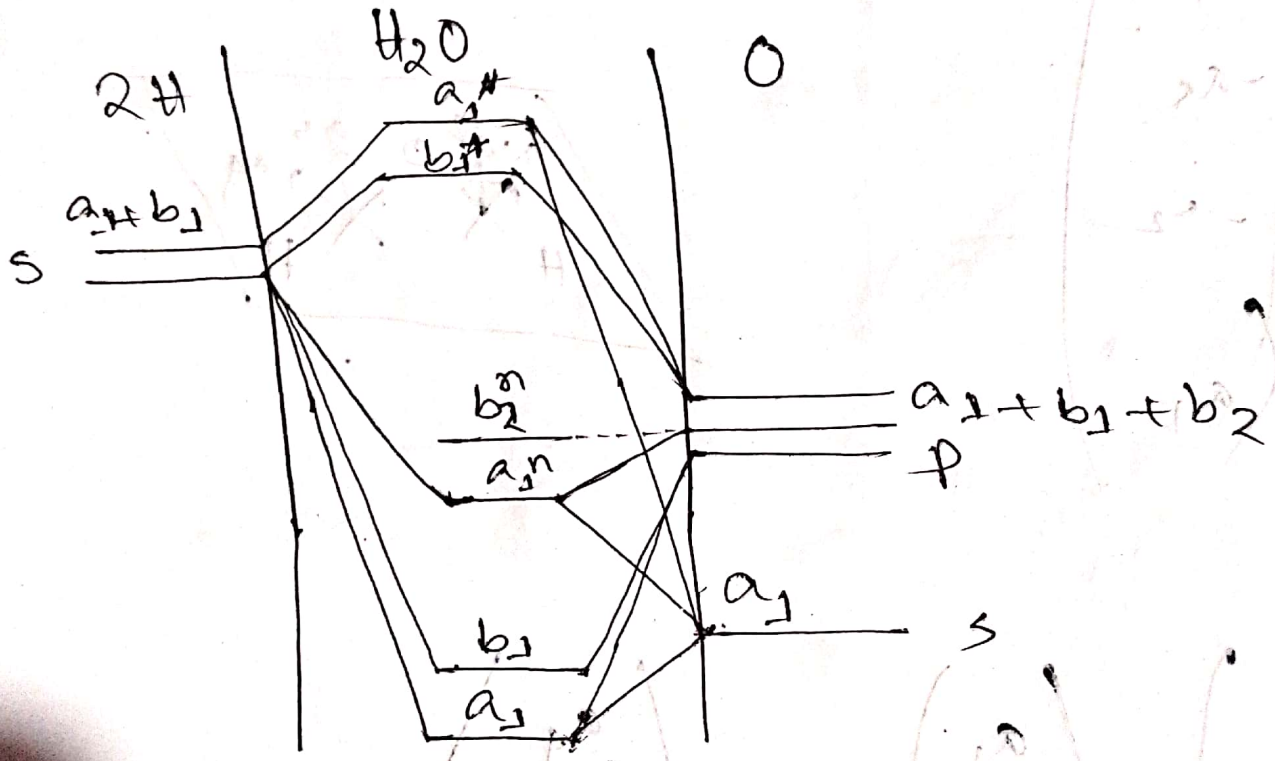


Here, O has 4 valance electron and 2H has 2 electron. We can put total $(4+2)=6$ electron into the molecular orbital.



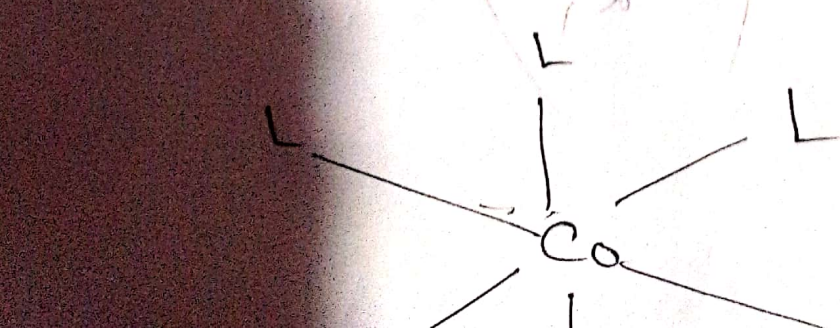
chapter-6

For H_2O



$Co(NH_3)_6^{3+}$

$\Gamma_7 = A_{1g} + E_g + T_{1u}$



* find the hybrid orbital of BCl_3 / BF_3

The complete set of character of D_{3h} is thus:

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	1	3	0	1

According to the characters table of D_{3h}:

No. of $A_1' = \frac{1}{12}[(3\times1\times1)+0+(1\times1\times3)+(3\times1\times1)+0+(1\times1\times3)]$

$= 1$

No. of $A_2' = \frac{1}{12}[(3\times1\times1)+0+(1\times(-1)\times3)+(3\times1\times1)+0+(1\times(-1)\times3]$

$= 0$

No. of $E' = \frac{1}{12}[(3\times2\times1)+0+(1\times0\times3)+(3\times2\times1)+0+(1\times0\times3)]$

$= 1$

No. of $A_1'' = \frac{1}{12}[(3\times1\times1)+0(1\times1\times3)+(3\times(-1)\times1)+0+(1\times(-1)\times3] = 0$

No. of $A_2'' = \frac{1}{12}[(3\times1\times1)+0+(1\times(-1)\times3)+(3\times-1\times1)+0+(1\times1\times3)] = 0$

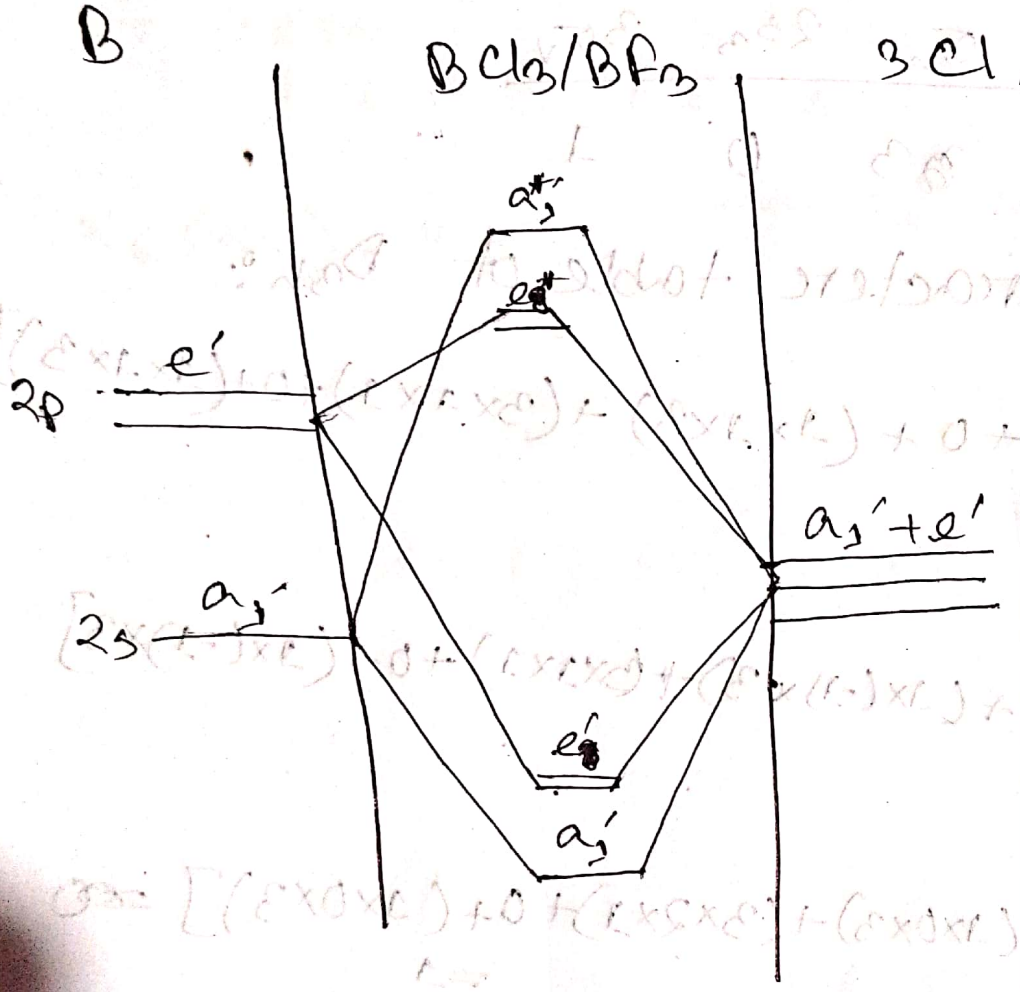
No. of $E'' = \frac{1}{12}[(3\times2\times1)+0+(1\times0\times3)+(3\times(-2)\times1)+0+(1\times0\times3) = 0$

So, we get, $\Gamma = A_1' + E'$

Here, we can see that, A_1' is belongs to dz^2 orbital or spherical s orbital. E' belongs to p_{xy}, p_{yz} orbital.

So, we get sp^2 orbital.

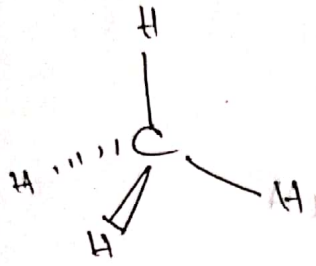
The correlation diagram of BCl_3 or BF_3.



For CH_4:

CH_4 is a tetrahedral molecule.

The structure of CH_4 is -



CH_4 belongs to T_d point group.

By taking 4 C-H bonds on the basis, we can calculate the groups orbital.

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

Now, considering the point group of T_d,

No. of $A_1 = \frac{1}{24}[(4\times1\times1)+(1\times1\times8)+0+0+(2\times1\times6)]$

$= 1$

No. of $A_2 = \frac{1}{24}[(4\times1\times1)+(1\times1\times8)+0+0+\{2\times(-1)\times6\}]$

$= 0$

No. of $E = \frac{1}{24}[(4\times2\times1)+\{1\times(-1)\times8\}+0+0+(2\times0\times6)]$

$= 0$

No. of $T_1 = \frac{1}{24}[(4\times3\times1)+\{(1\times0\times8)\}+0+0+\{2\times(-1)\times6\}]$

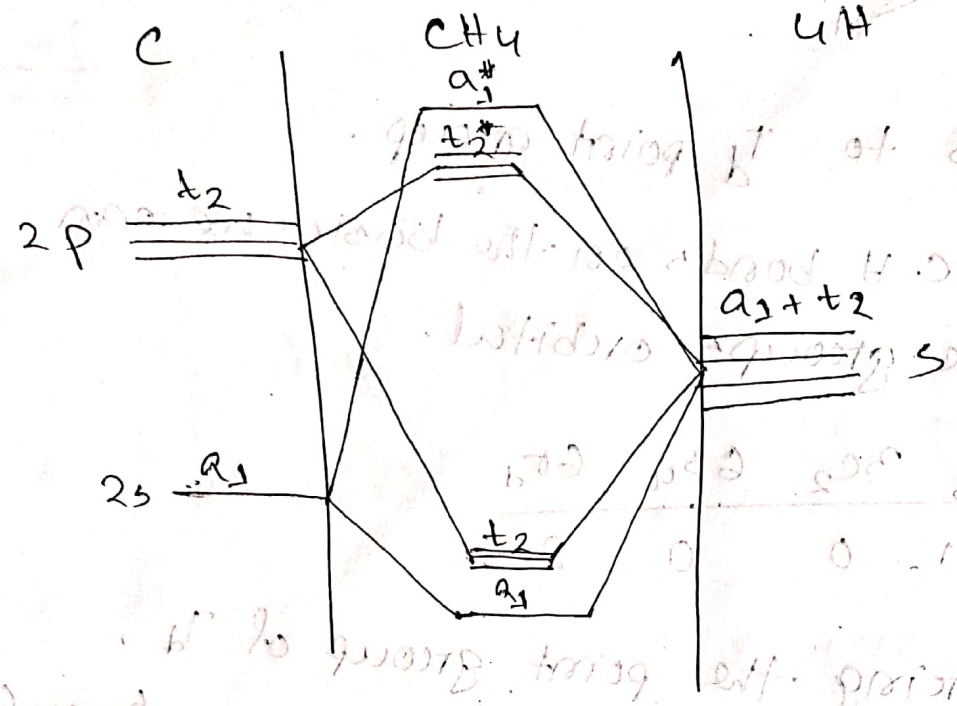
$= 0$

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No. of $T_2 = \frac{1}{24}[(4\times3\times1)+(1\times0\times8)+0+0+(2\times1\times6)]$

$= 1$

So, we get. $\Gamma = A_1 + T_2$.



Correlation diagram of CH_4.

Here, C has 4 valance electron, and 4 H atom have 4 electron. We can put total $(4+4) = 8$ electron into the molecular orbital

a_1^*, t_2^* } anti bonding

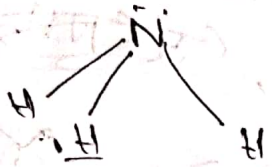
t_2, a_1 } bonding

A_{2u} - p_z

B_{2u} = none.

NH_3

The structure of NH_3 is trigonal pyramidal.



This belongs to point group C_{3v}.

2s orbital belongs to A_1 representation & 2Px and 2Py orbital E representation.

By taking three N–H bonds on, the basis, we can calculate the group orbitals for three H atom as follows.

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

No. of

No. of $A_1 = \frac{1}{6}\left[(3\times1\times1) + 0 + (-1\times1\times3)\right]$

$= 1$

No. of $A_2 = \frac{1}{6}\left[(3\times1\times1) + 0 + \{-1\times(-1)\times3\}\right]$

$= 0$

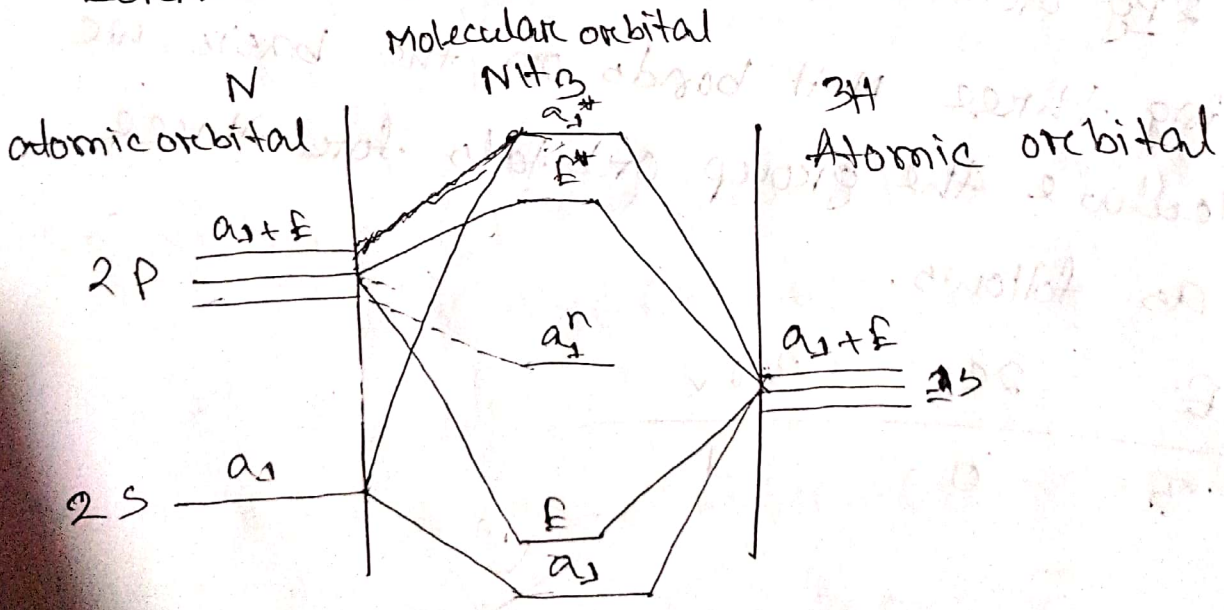
No. of $E = \frac{1}{6}\left[(3\times2\times1) + 0 + \{0\times(-1\times0\times3)\}\right]$

$= 1$

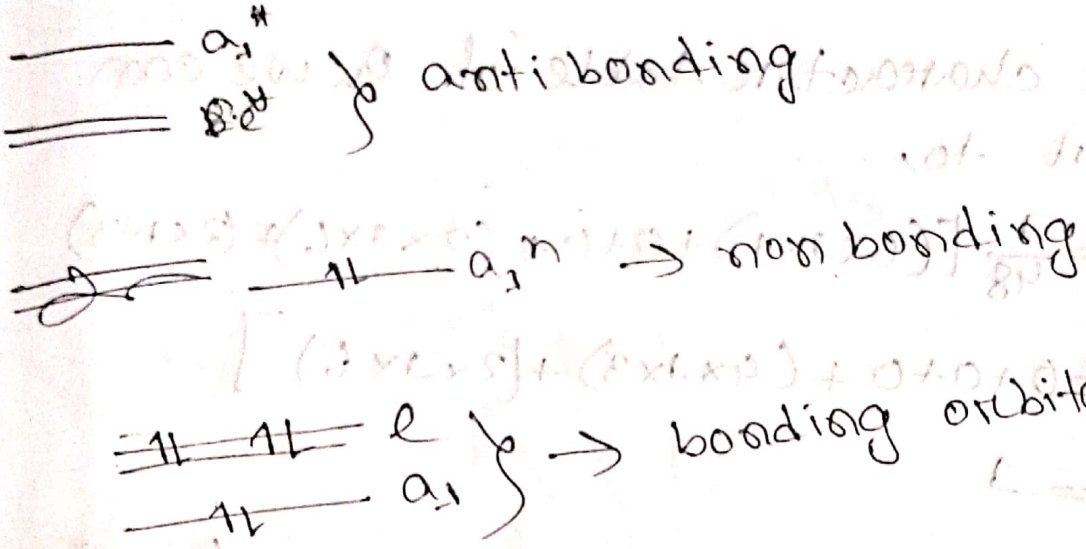
So, we get, $\Gamma = A_1 + E$

Chapter-6. H_2O, NH_3, $Co(NH_3)_6$, BCl_3, XeF_4, CH_4, $Co(NH_3)_6$

Correlation Diagram for NH_3:

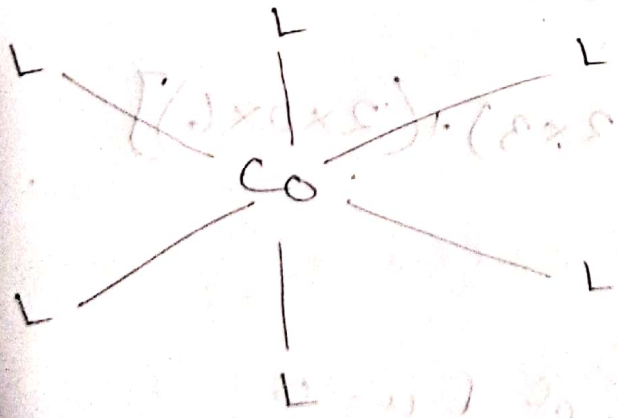


In NH_3 molecule, N has 5 valance electrons and there H atoms have 3 electron. Now we can put these 5+3 = 8 electrons into the molecular orbital as follows.



for $Co(NH_3)_6^{3+}$ molecule:

The structure of $Co(NH_3)_6^{3+}$ is a octahedreal molecule as follows:



by taking 6 $Co-NH_3$ bond on the basis we can calculate the group orbital:

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2(=C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$
	6	0	0	2	2	0	0	0	4	2

from the character table of O_h we can reduces it to,...

No. of $A_{1g} = \frac{1}{48}[(6\times1\times1)+0+0+(2\times1\times6)+(2\times1\times3)$
$+0+0+0+(4\times1\times3)+(2\times1\times6)]$
$= 1$

No. of $A_{2g} = \frac{1}{48}[(6\times1\times1)+0+0+(2\times(-1)\times6)+(2\times1\times3)$
$+0+0+0+(4\times1\times3)+(2\times(-1)\times6)]$
$= 0$

No. of $Eg = \frac{1}{48}[(6\times2\times1)+0+0+(2\times0\times6)+(2\times2\times3)$
$+0+0+0+(4\times2\times3)+(2\times0\times6)]$
$= 1$

No. of $T_{1g} = \frac{1}{48}[6\times3\times 0$

No. of $T_{2g} = 0$

No. of $A_{1u} = 0$

No. of $A_{2u} = 0$

No. of $Eu = 0$

No. of $T_{1u} = 1$

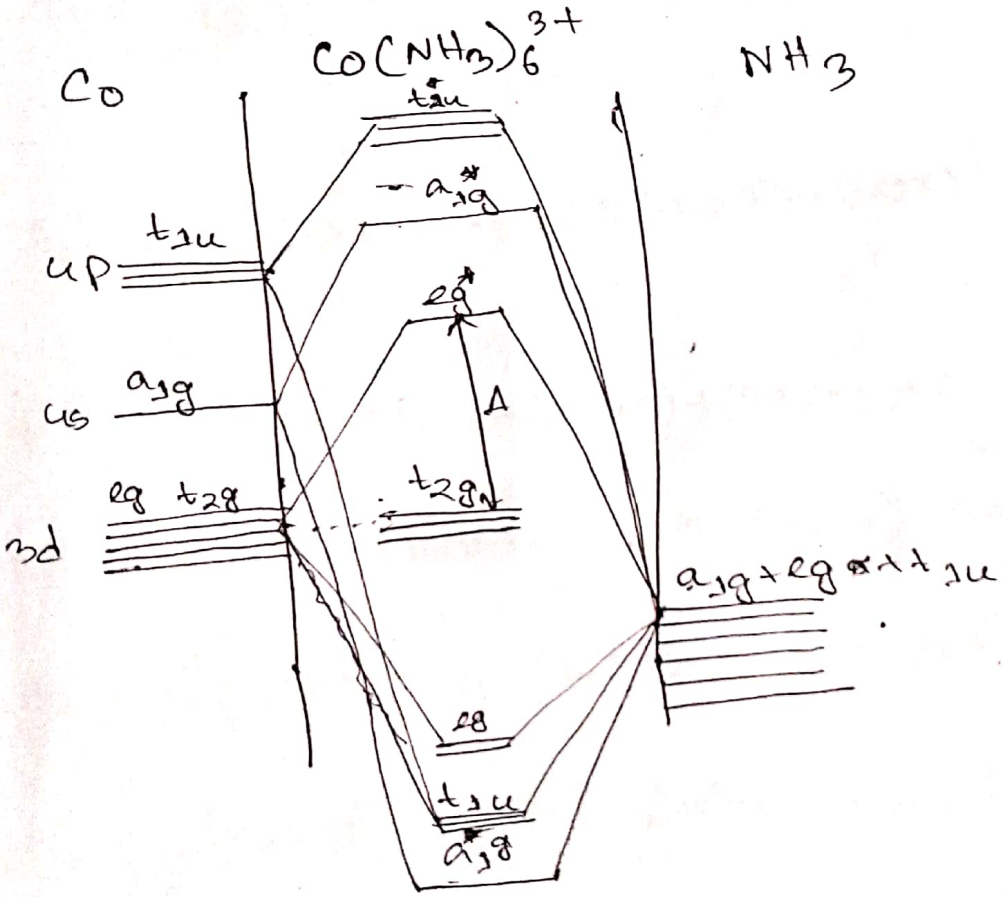
No. of $T_{2u} = 0$

$\therefore$ The reducible representation.

$\Gamma_T = A_{1g} + Eg + T_{1u}$.

The T_{2g} set of three metal ion orbitals.

The correlation diagram of $Co(NH_3)_6^{3+}$ is.

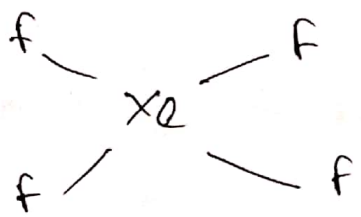


No. of $T_{2g} = \frac{1}{48}[(6\times3\times1)+0+0+(2\times(-1)\times6)+\{2\times(-1)\times3\}$
$+0+0+0+\{4\times(-1)\times3\}+(2\times1\times6)]$

$= \frac{1}{48}[18-12-6+0-12+12] = 0$

Q. Find the hybrid orbitals of a central atom suitable for forming a set of square planer bonds. Use the D_{4h} character table. [XeF_4]

→ A square planer bonds -



This molecule belongs to D_{4h} point group.

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	2	0	0	0	4	2	0

D_{4h} character table.

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
A_{1g}	1	1	1	1	1	1	1	1	1	1
A_{2g}	1	1	1	−1	−1	1	1	1	−1	−1
B_{1g}	1	−1	1	1						
B_{2g}	1	−1	1	−1						
E_g	2	0	−2	0						
A_{1u}	1	1	1	1						
A_{2u}	1	1	1	−1						
B_{1u}	1	−1	1	1						
B_{2u}	1	−1	1	−1						
E_u	2	0	−2	0						

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According to Dih character table:,

No. of $A_{1g} = \frac{1}{16}\{(4\times1\times1) + 0 + 0 + (2\times0\times2) + 0 + 0 + 0$

$+ (4\times1\times1) + (2\times1\times2)$

$= 1$

Same process,

No. of A_{2g} = ...

$= 0$

$B_{1g} = 1$ $\quad$ $A_{2u} = 0$

$B_{2g} = 0$ $\quad$ $B_{1u} = 0$

$Eg = 0$ $\quad$ $B_{2u} = 0$

$A_{1u} = 0$ $\quad$ $Eu = 1$

So, we can say that this reduces to:

$A_{1g} + B_{1g} + Eu$.

From the character table, we see that s (x^2+y^2) or dz^2 belongs to $\underline{A_{1g}}$ species.

dx^2-y^2 orbital belongs to $\underline{B_{1g}}$. Px, Py belongs to $\underline{Eu}$.

∴ A set of hybrid orbitals are dsp^2.

2. Find the orbitals suitable for 'out of plane' π-bonding in a square planar molecule [XeF$_4$]

→ The reducible representation of D_{4h} point group:

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	-4	0	0	-2	0	0	0	-4	2	0

According to D_{4h} character table:

No. of $A_{1g} = \frac{1}{16}\{(4\times1\times1)+0+0+\{(-2)\times1\times2\}+0+0+0$
$+\{(-4)\times1\times1\}+\{0\times1\times2\}$

$= 0$

No. of $A_{2g} = \frac{1}{16}[\{(4\times1\times1)+0+0\{(-2)\times1\times2\}+0+0+0$
$+\{(-4)\times1\times1\}+(2\times1\times2)]$

$= 0$

No. of $B_{1g} = \frac{1}{16}[\{(4\times1\times1)+0+0+\{(-2)\times(-1)\times2\}+0+0+0$
$+\{(-4)\times1\times1\}+\{(2\times1\times2)\}]$

$= 0$

No. of $B_{2g} = \frac{1}{16}[\{(4\times1\times1)\}+0+0+\{(-2)\times(-1)\times2\}$
$+0+0+0+\{(-4)\times1\times1\}+\{2\times(-1)\times2\}+0$

$= 0$

No. of $E_g = \frac{1}{16}[\{(4\times2\times1)\}+0+0+\{(-2)\times6\times2\}$

$+0+0+0+\{(-4)\times(-2)\times1\}+\{(2\times6\times3)\}$

$= 1$

No. of $A_{1u} = \frac{1}{16}[(4\times1\times1)+0+0+\{(-2)\times1\times2\}$

$+0+0+0+\{(-4)\times(-1)\times1\}+\{2\times(-1)\times2\}]$

$= 0$

No. of $A_{2u} = \frac{1}{16}[(4)\times1\times1+0+0+\{(-2)\times(-1)\times2\}$

$+0+0+0\{(-4)\times(-1)\times1\}+(2\times1\times32)]$

$= 1$

No. of $B_{1u} = \frac{1}{16}[(4\times1\times1)+0+0+\{(-2)\times1\times2\}+0+0+0$

$+\{(-4)\times(-1)\times1\}+\{2\times(-1)\times2\}]$

$= 0$

No. of $B_{2u} = \frac{1}{16}[(4\times1\times1)+0+0+\{(-2)\times(-1)\times2\}+0+0+($

$+\{(-4)\times(-1)\times1\}+(2\times1\times2)]$

$= 1$

No. of $E_u = \frac{1}{16}[(4\times2\times1)+0+0+\{(-2)\times0\times2\}+0+0+0$

$+\{(-4)\times2\times1\}+(2\times0\times2)]$

$= 0$

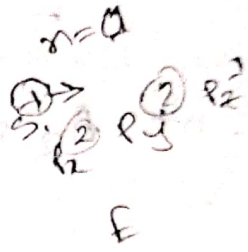
This representation reduces to $Eg + A_{2u} + B_{2u}$.

∴ suitable orbitals are -

Eg - d_{xz}, d_{yz}

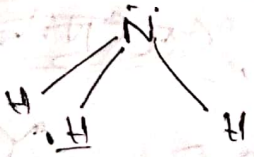
A_{2u} - p_z

B_{2u} = none.



NH_3

The structure of NH_3 is trigonal pyramidal.



This belongs to point group C_{3v}.

2s orbital belongs to A_1 representation ✓

2Px and 2Py orbital E representation.

By taking three N-H bonds on the basis, we can calculate the group orbitals for three H atom as follows.

The group orbitals:

1. H_2O (C_{2v})

C_{2v}	E	C_2	$\sigma_v(xy)$	$\sigma_v(yz)$
Γ	2	0	2	0

2. NH_3 (C_{3v})

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

3. BCl_3 / BF_3 (D_{3h})

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	1	3	0	1

4. XeF_4 (D_{4h})

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	2	0	0	0	4	2	0

5. CH_4 (T_d)

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

6. $Co(NH_3)_6^{3+}$ (O_h)

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2(=C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$
Γ	6	0	0	2	2	0	0	0	4	2

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